

Mohamed Mosaad

AI & Machine Learning Engineering

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🐙 GitHub 📁 Portfolio

SUMMARY

Passionate AI/Machine Learning Engineer with strong expertise in Deep Learning and practical experience in Computer Vision and NLP. Skilled in building advanced AI systems from Object Detection, Tracking, and Segmentation to Transformer-based NLP models and Agentic RAG systems using LangChain/LangGraph. Experienced in model development, fine-tuning, and backend integration (NestJS, Temporal). Constantly learning and applying state-of-the-art techniques through real-world projects shared on GitHub and Portfolio

EDUCATION

Tanta University, *AI Engineering* 10/2022 – 07/2027

EXPERIENCES

MESHKODE, *Junior AI Engineer - RAG & Agentic Systems* 07/2025 – 05/2026

- Designed and implemented Agentic RAG pipelines using LangChain and LangGraph, improving retrieval accuracy and response quality by 30%.
- Built scalable, fault-tolerant backend services using NestJS and Temporal, reducing processing latency and task failures by 25%

NTI, *Computer Vision & Deep Learning Training* 07/2025 – 08/2025

- Built and evaluated CV models using Python and OpenCV, achieving 90%+ accuracy
- Applied GANs and Grad-CAM to enhance model explainability and feature analysis by 30%

GTC, *Computer Vision & NLP Training* 08/2024 – 09/2024

- Completed intensive training in Computer Vision and Natural Language Processing (NLP) using Python and deep learning techniques
- Worked on image segmentation and NLP tasks, fine-tuning models to improve accuracy by up to 15% and reduce inference latency by 25%.

Elevo, *Machine Learning & Deep Learning Training* 06/2024 – 07/2024

- Completed hands-on training in Machine Learning and Deep Learning using real-world datasets
- Built and evaluated multiple models, achieving up to 98% accuracy across different tasks.

SKILLS

Languages (Python • C++ • TypeScript) | **Frameworks** (FastAPI • NestJS • LangChain • Lang Graph. Temporal) | **Library** (TensorFlow • PyTorch) | **Tools** (Git • GitHub • Hugging Face • Roboflow • Google Colab) | **Soft Skills** (Fast Learner • Problem Solving • Teamwork • Adaptability)

LANGUAGES

Arabic (Native) | **English** (Good)

PROJECTS

Multi-Agent & Code Intelligence Systems

- Built multi-agent system to analyze & refactor 1M+ LOC (Java, C#, C++, Python)
- Designed agents for parsing, reasoning, and code transformation using (LangChain / LangGraph / Temporal)
- Reduced manual engineering effort by ~50% in legacy systems

Blagh AI Platform (Multi-Agent RAG System)

- Built multi-role AI platform using Agentic RAG over 25GB domain data
- Designed 4 AI agents (public, scholar, admin, org-level) with controlled access
- Reduced latency by 20% and improved response quality

End-to-End RAG System with OCR

- Built fully local RAG using quantized LLaMA (7B)
- Fine-tuned Arabic OCRFlux 3B for document understanding
- Designed full pipeline (ingestion , embedding , retrieval , generation)
- integrated monitoring stack (Grafana) for performance tracking

Weapons Detection System

- Developed a real-time weapon detection system using YOLO.
- Achieved high detection accuracy in images and videos.
- Reduced inference latency by 30% for faster alerts.

Hajj & Umrah Fraud Detection

- Built multimodal pipeline (text + image) for fraud detection with 96% accuracy.
- Fine-tuned Arabic BERT for classification + explainability.